MODULAR HMS

Flask modular monolith for hospital workflows

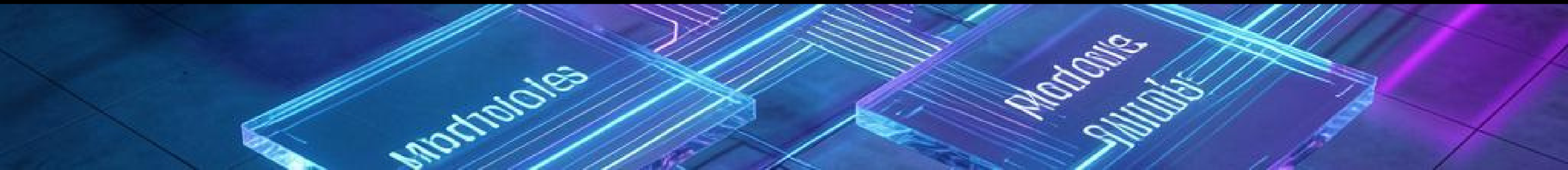
INTRODUCTION

A Python/Flask portal using Blueprints to separate auth, doctor, patient, and appointment logic.

Focus on role-based access and session security for a maintainable, scalable codebase.

01

ARCHITECTURE & MODULES



BLUEPRINT STRUCTURE

App split into distinct Blueprints to isolate responsibilities and enable parallel development.

init_db() runs at startup to ensure schema readiness and reduce runtime failures.

DATABASE INITIALIZATION

INIT_DB() RUNS AT STARTUP TO CREATE REQUIRED TABLES AND VERIFY SCHEMA.

DATABASE ACCESS IS HANDLED IN A SEPARATE MODELS LAYER TO KEEP ROUTING SIMPLE AND REDUCE RUNTIME ISSUES.

CONCLUSIONS

Modular Blueprints + init_db ensure a maintainable, reliable Flask monolith.

Session and role controls provide secure, role-specific access while supporting incremental scaling.

THANK YOU

Do you have any questions?